



**Ellen Ochoa**

1958–present

Ellen's skills as an engineer took her all the way into space.

After four exciting space missions, and completing almost 1,000 hours in orbit, she worked for NASA, becoming director of the Johnson Space Center.



**Max Planck**

1858–1947

Quantum theory is a set of ideas and laws that helps physicists describe and predict the behavior of the smallest particles in the universe. German physicist Max developed quantum theory in the early 1900s. Since then, it has led to discoveries and inventions that have changed the world.



**Claudius Ptolemy**

100–c.170 CE

For 1,400 years, the science of astronomy was based entirely on the ideas of the Egyptian scientist Ptolemy! His model of how the sun, the moon, and the planets orbit Earth turned out to be wrong, but it is important in understanding the history of science.



**Sally Ride**

1951–2012

Sally was a physicist and the first American woman to travel into space. She got the job after NASA advertised for young scientists to train as astronauts.

During two missions, Sally conducted dozens of different experiments, spending just over two weeks in space.



**Mashudu Tshifularo**

1964–present

In 1995, Mashudu became the first Black ear, nose and throat professor in South Africa. He later pioneered a new technique in ear surgery, using tiny replacement bones that had been made using 3-D printing technology.



**Irene Uchida**

1917–2013

Canadian scientist Irene built a huge genetic database of twins who were happy to help with research, allowing both her and other scientists to make important genetic discoveries.

Studying the similarities and differences between the genes of twins helps medical researchers understand conditions that have a genetic cause.



**Ian Wilmut**

1944–present

As a young biologist, Ian worked on genetic modification of animals to produce useful medicines. He is famous for leading the team that was the first to clone a mammal—a sheep named Dolly. Today, Ian works on improving the cloning of stem cells for medicine.



**Shinya Yamanaka**

1962–present

Stem cells promise to help us produce new types of medical treatment. Japanese researcher Shinya made a Nobel Prize-winning breakthrough when he developed a way of reprogramming ordinary body cells to turn them into stem cells.